

Data Model

Data preparation and feature engineering steps should be done in a separate project to match the expected input datasets format. Respecting the expected data model when loading new data helps ensure the results are reliable and easy to interpret.

Input datasets format

The input data should be separated into 5 different datasets:



revenues_data



product_holdings_data



customers_data



balances_data



additional_
information_data

The datasets revenues_data, product_holdings_data and balances_data must cover the **same period** and have the **same time frequency**.

- (1) revenues_data:

Revenue: the income a bank generates from each financial product it offers.

Dataset name	Description	Granularity	Mandatory
revenues_data	Revenues generated by each product per customer over time	1 Row = 1 transaction	yes ▾

revenues_data attributes				
Name	Description	Type	Example	Mandatory
date	Sequence of dates taken at successive equally spaced points in time (ex: quarterly, monthly)	date ▾	2021-03-01T00:00:00Z	yes ▾
product_id	Identification code of the product	string ▾	1201634630697007563	yes ▾
customer_id	Identification code of the customer	string ▾	182304	yes ▾
revenue	Revenue amount perceived	double ▾	10.3	yes ▾

- (2) product_holdings_data:

Dataset name	Description	Granularity	Mandatory
product_holdings_data	<i>Product information and duration period of each product held by customers</i>	<i>1 Row = 1 product held by a customer</i>	yes ▾

<i>product_holdings_data attributes</i>				
Name	Description	Type	Example	Mandatory
customer_id	Identification code of the customer	string ▾	182304	yes ▾
product_type	Product type name	string ▾	Secured Lending	yes ▾
product	Product name	string ▾	Car Loan	yes ▾
start_date	The date at which the subscription to the product started	date ▾	2015-01-28T00:00:00.000Z	yes ▾
product_id	Identification code of the product	string ▾	1201634630697007563	yes ▾
end_date	The date at which the subscription to the product ended or is supposed to end	date ▾	2015-04-28T00:00:00.000Z	yes ▾

Note : A product type is a product category that can encompass multiple individual products.

- (3) customers_data:

Dataset name	Description	Granularity	Mandatory
customers_data	<i>Customer static information dataset</i>	<i>1 Row = 1 customer</i>	yes ▾

<i>product_holdings_data attributes</i>				
Name	Description	Type	Example	Mandatory
customer_id	Identification code of the customer	string ▾	182304	yes ▾
creation_date	Creation date of the account by the customer	date ▾	2021-03-01T00:00:00Z	yes ▾
tier	Categorization of customers	string ▾	GOLD	yes ▾
birth_date	Birthdate of the customer	date ▾	1956-01-28T00:00:00.000Z	yes ▾
first_name	First name of the customer	string ▾	Niccolo	yes ▾
last_name	Last name of the customer	string ▾	Ryles	yes ▾
optional_col1	Optional column to add in the dataset	anyT... ▾	ex: Female	no ▾
...	...	anyT... ▾	...	no ▾
optional_coln	Optional column to add in the dataset	anyT... ▾	ex: 123902	no ▾

- (4) balances_data:

Balance: money held by the client (like in a checking account) or owed by the client (like a credit card or mortgage).

Dataset name	Description	Granularity	Mandatory
balances_data	<i>Balance amounts of each product per customer over time</i>	<i>1 Row = 1 balance at a specific date</i>	yes ▾

<i>revenues_data attributes</i>				
Name	Description	Type	Example	Mandatory
date	Sequence of dates taken at successive equally spaced points in time (ex: quarterly, monthly)	date ▾	2021-03-01T00:00:00Z	yes ▾
product_id	Identification code of the product	string ▾	1201634630697007563	yes ▾
customer_id	Identification code of the customer	string ▾	182304	yes ▾
balance	Balance amount at this time	double ▾	10.3	yes ▾

- (5) additional_information_data:

Dataset name	Description	Granularity	Mandatory
additional_information_data	<i>Additional information about the customer through time</i>	<i>1 Row = 1 evenment at a specific date</i>	yes ▾

<i>revenues_data attributes</i>				
Name	Description	Type	Example	Mandatory
date	Sequence of dates taken at successive equally spaced points in time (ex: quarterly, monthly)	date ▾	2021-03-01T00:00:00Z	yes ▾
customer_id	Identification code of the customer	string ▾	182304	yes ▾
optional_col1	Optional column to add in the dataset	anyT... ▾	ex: In person interaction	no ▾
...	...	anyT... ▾	...	no ▾
optional_coln	Optional column to add in the dataset	anyT... ▾	ex: ATM withdrawals	no ▾

The datasets containing revenue, balance, and product holding information include only mandatory columns. In contrast, the datasets for customers and additional information may be enriched with optional columns that can enhance the classification model's performance.